

QUAKERTOWN, PA, USA

WMO: 720324

Lat: 40.435N

Lon: 75.382W

Elev: 160

StdP: 99.41

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 64753

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.8	-11.3	-21.0	0.6	-12.3	-17.9	0.8	-8.9	8.3	-1.0	7.5	-0.3	1.8	310	0.428

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	32.6	23.6	31.3	23.0	29.9	22.3	25.2	30.2	24.3	28.6	23.5	27.5	2.5	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.8	19.0	27.0	22.9	18.0	26.0	22.5	17.5	25.5	77.8	29.5	74.1	28.4	71.1	27.7	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.3	6.0	5.3	DB	-16.8	35.5	3.2	1.9	-19.0	36.8	-20.9	37.9	-22.7	39.0	-25.0	40.4	(4)
(5)			WB	-17.9	26.7	3.1	0.7	-20.2	27.1	-22.0	27.5	-23.7	27.9	-26.0	28.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.1	-1.4	-1.1	3.9	10.5	15.9	20.7	23.5	22.3	18.5	12.3	5.8	1.4
	DBStd	9.82	5.45	5.15	5.16	4.72	4.40	3.31	2.74	2.86	3.73	4.62	4.47	4.87
	HDD10.0	1358	354	311	200	49	4	0	0	0	0	28	140	271
	HDD18.3	3106	612	543	448	239	103	16	1	3	43	194	376	526
	CDD10.0	1756	1	1	10	64	187	320	419	380	256	100	13	4
	CDD18.3	462	0	0	0	4	28	86	162	125	49	8	0	0
	CDH23.3	4256	0	0	3	74	317	787	1524	1081	419	50	0	0
	CDH26.7	1439	0	0	0	17	92	244	591	362	124	9	0	0
Wind	WSAvg	1.7	1.8	2.1	2.5	2.4	1.7	1.5	1.2	1.0	1.2	1.6	1.8	1.8
Precipitation	PrecAvg	1221	91	66	107	96	99	107	124	107	118	114	84	109
	PrecMax	1602	164	163	169	227	173	265	267	221	278	275	144	248
	PrecMin	980	39	22	24	30	32	35	18	37	11	6	30	31
	PrecStd	154	38	31	38	46	37	54	52	42	67	69	29	48
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.2	17.8	22.8	28.8	32.0	33.6	35.0	33.9	32.6	27.6	20.9	17.9
		MCWB	12.9	13.1	15.0	17.0	21.9	24.0	24.4	24.1	23.4	20.5	16.2	16.0
	2%	DB	12.6	13.0	18.7	25.2	29.1	31.3	32.8	32.2	30.0	24.1	18.0	14.0
		MCWB	10.5	8.6	13.0	15.9	21.0	22.9	23.8	23.8	21.9	19.0	14.1	12.3
	5%	DB	8.8	9.0	14.9	22.5	27.3	29.0	32.0	30.1	27.6	22.4	16.1	12.0
		MCWB	6.8	5.6	9.8	14.6	19.6	21.9	23.3	22.7	20.6	18.0	12.1	10.6
	10%	DB	7.1	6.3	12.4	19.2	24.2	27.7	30.0	28.7	25.9	20.1	13.7	8.8
		MCWB	5.0	3.5	8.5	12.8	17.9	20.6	22.5	21.9	19.6	16.7	10.6	6.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.5	14.9	16.6	19.7	23.9	25.0	26.5	26.1	25.1	22.8	17.4	16.5
		MCDB	15.2	17.7	21.0	23.1	28.6	31.0	32.2	31.9	29.6	25.7	19.3	17.6
	2%	WB	10.9	9.8	13.7	17.6	22.2	24.0	25.4	25.1	23.6	21.2	15.4	13.0
		MCDB	12.3	14.1	17.3	22.7	27.4	28.8	30.5	29.9	26.9	23.0	17.6	13.8
	5%	WB	7.4	6.0	11.1	15.8	20.9	23.0	24.5	24.0	22.5	19.4	13.1	10.2
		MCDB	8.2	8.5	14.0	20.9	25.7	27.5	28.9	27.7	25.6	21.8	15.1	11.1
	10%	WB	4.8	3.5	8.6	14.0	19.5	22.2	23.7	23.0	21.4	17.3	11.3	7.3
		MCDB	6.5	6.0	11.8	18.4	23.1	26.2	27.8	26.4	24.4	18.8	13.2	8.2
Mean Daily Temperature Range	5% DB	MDBR	8.6	9.4	10.3	12.2	12.1	11.3	11.4	11.3	11.7	10.8	10.2	8.2
		MCDBR	12.7	13.9	15.0	16.8	15.1	13.5	13.3	12.7	13.9	13.5	13.7	11.3
		MCWBR	11.4	10.5	10.3	8.9	7.5	6.0	5.1	5.3	6.7	7.9	9.8	10.4
	5% WB	MCDBR	11.7	12.9	13.3	13.6	13.1	11.6	10.7	10.4	11.1	9.6	11.0	10.3
		MCWBR	11.0	10.5	10.8	8.8	7.1	5.9	4.8	4.9	5.9	6.9	9.3	10.4
Clear-Sky Solar Irradiance	taub		0.303	0.319	0.355	0.385	0.428	0.487	0.486	0.481	0.396	0.356	0.322	0.303
	taud		2.453	2.390	2.369	2.312	2.221	2.096	2.113	2.113	2.322	2.447	2.505	2.504
	Ebn at Noon		860	895	896	888	852	801	797	788	841	842	833	834
	Edh at Noon		78	96	110	125	140	159	155	151	115	91	75	70
All-Sky Solar Radiation	RadAvg		1.84	2.74	3.63	4.72	5.26	5.74	5.91	5.11	4.24	2.98	2.09	1.57
	RadStd		0.16	0.31	0.28	0.45	0.62	0.48	0.30	0.33	0.53	0.32	0.19	0.17

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (68 neighbors)		+0.41	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+120	+68	

Nomenclature: See separate page